CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2012

EC210 Analog Communication

Date: 18.05.2012, Friday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define Fourier transform and Inverse Fourier Transform. State all properties of Fourier transform and derive any three. [05]

(b) Compare power signal and energy signal. [02]

Q - 2 (a) Explain diagonal peak clipping and negative peak clipping in the diode envelope detector with suitable circuit and waveforms. [08]

(b) The antenna current of an AM broadcast transmitter, modulated to a depth of 40 percent by an audio sine wave, is 11A. It increases to 12 A as a result of simultaneous modulation by another audio sine wave. What is the modulation index due to this second wave? [06]

OR

Q - 2 (a) Derive the mathematical equation of average power, effective voltage and effective current for sinusoidal and non sinusoidal amplitude modulation. [07]

(b) The antenna current of an AM transmitter is 8ampere (8A) when only the carrier is sent, but it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percent of modulation changes to 0.8. [07]

Q - 3 (a) Give the disadvantages of balance modulator-filter method. How it overcome with next SSB (single sideband) generation? [02]

(b) Draw the block diagram of SSB pilot carrier radio system with transmitter and receiver and explain it. [07]

(c) Differentiate between amplitude modulation (AM), frequency modulation (FM) and Phase modulation (PM) with is waveform. [05]

OR

Q - 3 (a) Draw the amplitude modulated signal spectra of following system [02]

1)Double sideband full carrier(DSBFC)

2) Double sideband suppressed carrier(DSBSC)

3) Single sideband suppressed carrier(SSBSC)using upper sideband (USB)

4) Single sideband suppressed carrier(SSBSC)using lower sideband (LSB)

(b) Write a note on doubly balanced diode ring modulator. [07]

(c) Write a note on Armstrong method for frequency modulation (FM). [05]

SECTION – II

Q - 4 (a) Explain the concept of pre-emphasis and de-emphasis with suitable circuit and frequency response. [05]

(b) Write mathematical equation for modulation index of frequency modulation (FM) and mention the value for wideband ,narrowband FM. [02]

Q - 5 (a) Draw the block diagram of FM broadcast receiver and explain it. [05]

(b) Explain the concept of sensitivity and gain with suitable voltage levels at various point of receiver. [06]

(c) Define-frequency tuning range ratio, capacitance tuning range ratio [03]

OR

Q - 5 (a) Explain foster-seeley discriminator with its circuit and waveforms. [05]

(b) A receiver tunes signals from 550 to 1600 KHz with an IF of 455KHz.find the frequency tuning ranges and capacitor tuning ranges for the oscillator section and for the RF section. [06]

(c) Define-image frequency, double spotting [03]

Q - 6 (a) What is tracking in receiver? Give its different types. [03]

(b) Calculate the noise voltage at the input of a TV RF amplifier, using a device that has a 200Ω equivalent noise resistance and a 300Ω input resistor. The bandwidth of the amplifier is 6MHz, and the temperature is 17°C. [06]

(c) Explain the following terms. 1)Shot noise 2)Johnson noise [05]

OR

Q – 6 (a) Why double conversion receiver required at higher frequency? [03]

(b) A receiver connected to an antenna whose resistance is 50Ω has an equivalent noise resistance of 30Ω.Calculate the receiver's noise figure in decibels and its equivalent noise temperature. [06]

(c) Explain the following terms. 1)Flicker noise 2)Burst noise [05]

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